FLEET TEAM DIGEST

737MAX-FTD-27-18003

Issue Title : Leading Edge Flaps/Slats Proximity Sensor Reliability Study

Airplane Model	ATA	Minor Model(s)
737MAX	2781-00	-8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
09/19/2018	08/01/2025	09/30/2024

Estimated Completion Date	Next Update Date

Revision Description

Provided updates to the Milestone Section for release.

Applicability

This issue is applicable to all 737NG/MAX airplanes equipped with proximity sensor PN: 80-207-01/02 (Boeing PN: 10-61226-45/47).

Description

Operators have experienced Air Turn Backs (ATBs) and dispatch interruptions due to Leading Edge (LE) Flap/Slat Fault Indications. For several operators, the LE Flap/Slat Indication System is considered as one of the top 10 drivers in their Fleet Dispatch Interruptions. Some of these events occurred within the first 2 years of aircraft operation, after approximately 4,800 FH / 3,800 FC.

As a result, a LE Flap/Slat Proximity Sensor Reliability Investigation was initiated in June 2018 on removed sensors PN: 80-207-02 which found two failure mechanisms:

1) Magnetic Set of Cores

2) Temperature Compensating (TC) Resistor Failure

A proximity sensor affected by the Magnetic Set of Cores failure condition is a sensor that has been magnetized and has reached its saturation point, thus causing a decreased sensor gap actuation performance. The effect observed on this proximity sensor is poor target NEAR performance and will continue to decrease until the issue is identified and addressed.

A proximity sensor affected by the TC Resistor failure condition is a sensor that has experienced moisture ingression from the ground terminal (airplane side) and has wicked into the sensor's ground cable (yellow wire). Over time, this moisture causes internal component corrosion where a conductive path is created. The effect observed on this sensor is lowered resistance measurements across the R1 connection (Yellow to Blue).

See Background for recent sensor investigation results.

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Background

Boeing and CAE are working together to provide solutions to the following failure mechanisms:

Magnetic Set of Cores

Failure Mode: Repeat or Intermittent System Level Indications from Leading Edge Flaps/Slats System

Root Cause: Magnetized proximity sensor reaches saturation and causes reduced sensor's actuation gap performance.

Possible Causes of a Magnetically Set Core include:

1) Target Sensor Contact (Can Cause Magnetic Set)

2) Exposing the sensor and possibly wiring to a strong electro-magnetic field

3) Placing a magnetic tip screwdriver on the sensing surface

4) Placing a metal target on the sensing surface at Zero gap when troubleshooting

5) Exposing the sensing surface to power tools

6) Side Metal Effects (exposure to magnetized mounting brackets, bolts & screws)

TC Resistor Failure

Failure Mode: Repeat or Intermittent System Level Indications from Leading Edge Flaps/Slats installation

Root Cause: Moisture ingress from the ground terminal into the sensor (yellow wire)

Possible Causes of TC Resistor Failures include:

1) Ground terminal connection not sealed with protective sealant

2) Moisture ingress leads to internal component corrosion

3) Corroded TC Resistor creates a conductive path and results to a low resistance measurement across the (Yellow to Blue) connection

Status

Magnetic Set of Cores

Proposed Solutions:

1) Introduce Trend Monitoring Program to perform proactive Inductance & Resistance Measurements via an LCR Meter with or

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without a Flaps Slats Electronics Unit (FSEU) Breakout Box. MAY-2019 (complete)

a) CAE completed automation of data logging process to obtain inductance and resistance measurements. MAR-2019 (complete)

2) CAE completed evaluation and prototype design of an on-wing degaussing test equipment. MAR-2019 (complete)

TC Resistor Failure

Proposed Solutions:

1) Introduce Trend Monitoring Program to perform proactive Inductance & Resistance Measurements via an LCR Meter with or without an FSEU Breakout Box. NOV-2019 (target)(complete)

a) CAE completed automated data logging process to obtain inductance and resistance measurements. MAR-2019 (complete)

2) Boeing's Ground Terminal Wiring Protection Project: Cancelled because CAE's Sensor Wiring Protection Project was determined to be adequate NOV-2023
Boeing is evaluating sealing methods for wire ground terminals that are exposed to moisture in the wing leading edge.
a) Seal all environmentally exposed ground connections within the LE Flaps/Slats Location

3) CAE's Sensor Wiring Protection Project: MAR-2022 (complete)

a) Add conformal coating on specific internal sensor components using (HumiSeal 1B73) material

b) Wick in a low viscous material on the wires to be crimped using material

Interim Action

Fleet Support Solution:

1) Institute trend monitoring program (12-14 Month Basis)

2) Employ ground terminal sealing practice under 737-SL-24-207 into operator maintenance programs

3) Utilize proximity sensor best practices under 737-SL-32-196 into operator maintenance programs

Final Action

Incorporation of new sensor on MAX production line - OCT-2021(complete)

Milestones

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Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: DEC-2018 (complete)

Solution Selected: FEB-2019 (complete)

Production Incorporation: Date - OCT-2021(complete)

Boeing Service Letter (737-SL-27-303): OCT-2024 (complete)

Supplier Service Letter: MAR-2022 (complete)

Operator Action

For Magnetic Set of Cores & TC Resistor Failure:

1) Maintenance Action: During troubleshooting, utilize proximity sensor best practices under 737-SL-32-196 aside from performing the isolation steps in the respective FIMs for specific FDE, fault message or maintenance message number.

2) Maintenance Action: If not already done so, seal in ground terminal connection using 737-SL-24-207 as a guidance. This is the yellow wire coming from the sensor and terminating at the ground terminal on-wing.

Supplier Information

PN: 80-207-01 (10-61226-45)

PN: 80-207-02 (10-61126-47)

Crane Aerospace & Electronics

16700 13th Avenue West

Lynnwood Washington, 98037, USA

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Letter (SL)	SL-737-32-196	B	01/26/2018
Service Letter (SL)	SL-737-24-207	A	08/16/2016

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Aircraft Maintenance Manual (AMM)	27-88-00	66	06/15/2018
Fault Isolation Manual (FIM)	27-88-00	66	06/15/2018
Service Letter (SL)	737-SL-27-303		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Proximity Sensor	10-61226-45	Crane Aerospace & Electronics	80-207-01
Proximity Sensor	10-61226-47	Crane Aerospace & Electronics	80-207-02

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- AOGs : Events that cause a significant Airplane On Ground (AOG) condition
- Best Practices / Operational Efficiency : Operational / Maintenance-focused best practices, techniques, health checks or recommendations for preserving aircraft efficiency.
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations
- Fleet Team IdeaXchange : Topic of discussion on the Fleet Team IdeaXchange bulletin board

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